

Cheng Zhen

Provost's Distinguished Graduate Fellow | Ph.D. Candidate in Computer Science at Oregon State University

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RESEARCH INTERESTS

My research primarily focuses on data systems for artificial intelligence, with a special emphasis on scalable data cleaning and data quality evaluation to efficiently prepare data for training large AI models. My recent work also studies machine learning over incomplete data, LLM evaluation, AI agents, and automated data/model preparation pipelines.

EDUCATION

Ph.D. in Computer Science

Anticipated Graduation: June 2027

Oregon State University, Corvallis, OR | GPA: 3.9/4.0

- Selected coursework: Database Management, Natural Language Processing, Convex Optimization

Master of Electrical and Computer Engineering

Jun 2022

Oregon State University, Corvallis, OR | GPA: 3.9/4.0

- Selected coursework: Computer Architecture, Nonlinear Optimization

Bachelor of Materials Science and Engineering

Jul 2019

Northwestern Polytechnical University, Xi'an, China

- Graduated with honors. Minor in Law.

RESEARCH EXPERIENCE

Graduate Research Assistant

Jun 2022 - Present

"Scalable Data Preparation for Large AI Models"

Oregon State University, Corvallis, OR

- Developing automated pipelines for data cleaning and augmentation, efficiently preparing large-scale datasets from minimal raw data for model training
- Achieved ~30% cost reduction in data preparation by identifying unnecessary cleaning operations and uninformative augmentations
- Applying automated systems for training large AI models including deep neural networks
- Collaborating with NEC Labs America and Intel Labs via the NSF Pervasive Personalized Intelligence Center

Provost's Distinguished Graduate Fellow

Mar 2020 - Jun 2022

"NASA Early-Stage Innovation Project"

Oregon State University, Corvallis, OR

- Developed generative models using deep neural networks for designing large-scale optical devices
- Built and optimized a retrieval-augmented generation module that integrates an external electromagnetic solver, enhancing reliability and doubling generation efficiency
- Worked in collaboration with teams from NASA and the Georgia Institute of Technology

Research Intern

Sep 2018 - Feb 2019

"Machine Learning for Multi-layer Optics"

Sorbonne University, Paris, France

- Estimated parameters for optical devices using least square approximation with a low estimation error of 2%
- Enhanced visualization of a large-scale multi-dimensional optical spectrum using Python

INDUSTRY EXPERIENCE

Ph.D. Software Engineer Intern

Aug 2025 - Dec 2025

Google, Kirkland, WA

- Built an end-to-end agent evaluation pipeline with data generation, simulation, and loss categorization; 3x evaluation efficiency
- Extended into a closed-loop agent improvement pipeline, improving performance without curated datasets
- Member of the Gemini Quality & Tooling team, contributing to an internal Gemini agent evaluation benchmark

AI Scientist Intern

Jun 2025 - Aug 2025

Salesforce, San Francisco, CA

- Developed an LLM-based multi-agent system to iteratively optimize deep learning training codebase with minimal supervision
- Enhanced context management across agents, enabling automated code revision and benchmarking in a closed-loop framework

Research Intern

Nov 2024 - Apr 2025

Samsung Research America, Mountain View, CA

- Developed a synthetic data generation pipeline for automatically evaluating the performance of AI agents in healthcare domains
- Enhanced an LLM-based evaluation framework by automatically optimizing prompts with minimal human labeling efforts
- Led a research paper accepted to ACL 2025 (Industry Track) and contributed to a related patent application

PUBLICATIONS

Manuscripts, Data Systems, and AI

1. **Cheng Zhen**, et al.. "A Proposer-Solver-Verifier Framework for Co-Evolutionary Agent Improvement." *Pending Internal Review*.
2. **Cheng Zhen**, Prayoga FNUP, Nischal Aryal, Arash Termehchy, and Alireza Aghasi. "Minimal Data Cleaning for Model Training by MinPrep." *PVLDB 2026 / VLDB 2026 Demonstration Track*, *accepted*.
3. Cheng Zhen, Nischal Aryal, Arash Termehchy, Prayoga, Garrett Biwer, and Sankalp Patil. "Learning Accurate Models on Incomplete Data with Minimal Imputation." arXiv:2503.13921 [cs.LG], 2025.
4. Cheng Zhen, Nischal Aryal, Arash Termehchy, Prayoga, and Garrett Biwer. "Minimal Repairs for Learning Over Incomplete Data." *NeurIPS Workshop on Reliable ML from Unreliable Data*, 2025.
5. Cheng Zhen, Ervine Zheng, Jilong Kuang, and Geoffrey Jay Tso. "Enhancing LLM-as-a-Judge through Active-Sampling-based Prompt Optimization." *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL Industry Track)*, 2025.
6. Cheng Zhen, Nischal Aryal, Arash Termehchy, Alireza Aghasi, and Amandeep Singh Chabada. "Certain and Approximately Certain Models for Statistical Learning." *Proceedings of the ACM on Management of Data (SIGMOD)*, Article 126, 2024.
7. Cheng Zhen, Amandeep Singh Chabada, and Arash Termehchy. "When Can We Ignore Missing Data in Model Training?." *Proceedings of the SIGMOD Workshop on Data Management for End-to-End Machine Learning (DEEM)*, 2023; also presented at *SDM 2024 workshop*.

Materials Science, Photonics, and Sensing

8. Boxin Zhang, Xingwei Hou, Cheng Zhen, and Alan X. Wang. "Sub-Part-Per-Billion Level Sensing of Fentanyl Residues from Wastewater Using Portable Surface-Enhanced Raman Scattering Sensing." *Biosensors* 11(10):370, 2021.

9. Wei-Che Hsu, Cheng Zhen, and Alan X. Wang. "Electrically Tunable High-Quality Factor Silicon Microring Resonator Gated by High Mobility Conductive Oxide." ACS Photonics 8(7):1933-1936, 2021.
10. Wei-Che Hsu, Cheng Zhen, and Alan X. Wang. "Tunable High Quality-Factor Silicon Microring Resonator Driven by High-Mobility Transparent Conductive Oxide." Conference on Lasers and Electro-Optics (CLEO), 2021.
11. Wei-Che Hsu, Bokun Zhou, Cheng Zhen, and Alan X. Wang. "Silicon Microring Resonator Driven by High-Mobility Conductive Oxide Capacitor." IEEE Photonics Conference (IPC), 2020.
12. Mack H. Cullison, Cheng Zhen, Jeshua J. Olson-Gross, Yi Xie, Michael T. Benson, and Tianyi Chen. "Microstructure Study of U-35 wt.% Zr Alloy after Quick Annealing at 650 °C." Journal of Materials Research 35(8):1095-1102, 2020.
13. Zhixin Li, Xiaoqiang Li, Boxin Zhang, Cheng Zhen, et al.. "Enhanced Thermal and Mechanical Properties of Optimized SiCf/SiC Composites with In-Situ CNTs on PyC Interface." Ceramics International 46(11):18071-18078, 2020.
14. Cheng Zhen, Xiaoqiang Li, Boxin Zhang, Xianguo Zhou, Chuanxin Liu, Zhixin Li, Yu Jiang, Litong Zhang, and Laifei Cheng. "The Improvement of Mechanical Properties of SiC/SiC Composites by In Situ Introducing Vertically Aligned Carbon Nanotubes on the PyC Interface." Ceramics International 45(3):3368-3376, 2019.

HONORS AND AWARDS

- Provost's Distinguished Graduate Fellowship, Oregon State University (top 1%)
- Most Impactful Research Award, National Science Foundation - Pervasive Personalized Intelligence Center (top 2%)
- CHIP Award, National Science Foundation
- Professional Development Award (twice), Oregon State University
- Scholarly Presentation Award, Oregon State University
- National Scholarship, The Ministry of Education, China (top 1%)
- Outstanding Delegation Award, National Model United Nations Conference - New York

MENTORING

- Mentored master's students Amandeep Singh Chabada and Prayoga FNUP on research projects in data preparation, machine learning, and data systems
- Mentored undergraduate students Sankalp Patil and Garrett Biwer on research related to building reliable data systems

TEACHING EXPERIENCE

CS540: Database Management Systems | CS462/463: Senior Software Engineering Project | CS 467: Capstone Project | ENGR 201/202: Electrical Fundamentals | ECE 582: Optical Electronic Systems | ECE 271: Digital Logic Design

- TA evaluation result: exceeding expectations

ACADEMIC SERVICE

Reviewer for ACL 2026, VLDB 2024, ICML 2024, ICDE 2023, and EDBT 2023